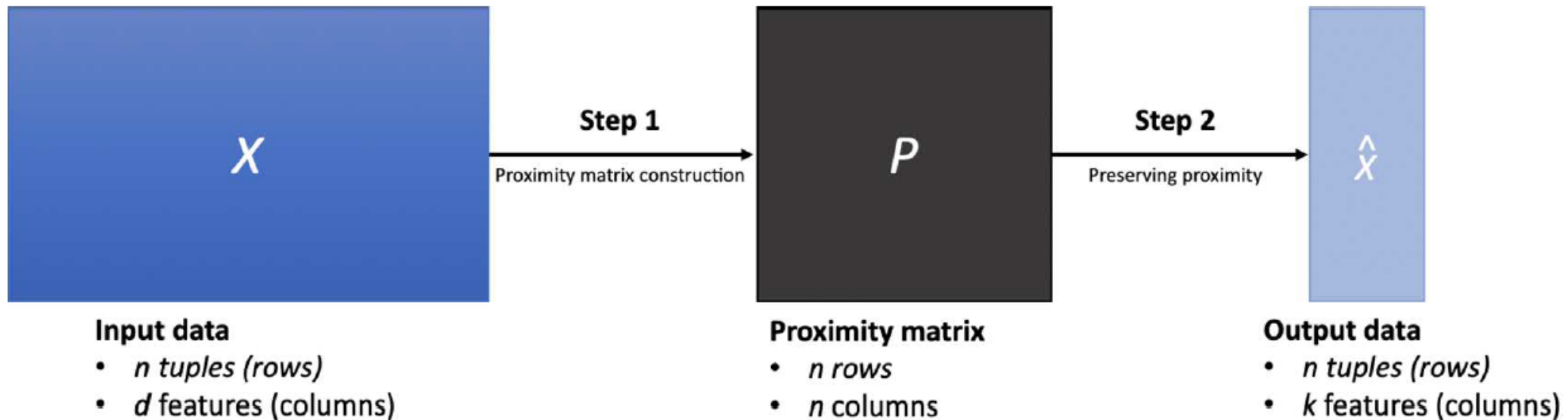


Nonlinear Dimensionality Reduction Methods

- PCA is a linear method for dimensionality reduction
 - Each principal component is a linear combination of the original input attributes
 - It works well if the input data approximately follows a Gaussian distribution or forms a few linearly separable clusters
- When the input data is linearly inseparable, we need to construct a proximity matrix (P) and learn a new matrix with k features ($k \ll d$) that preserves the proximity



Nonlinear Dimensionality Reduction (I): Kernel PCA (KPCA)

- ❑ Use a kernel function $\kappa(\cdot)$ to construct the kernel matrix: $P(i, j) = \kappa(x_i, x_j)$, and learn the best low-dimensional representations so that the estimated proximity matrix $\hat{P}$ is as close as possible to the kernel matrix P
- ❑ This can be obtained by using top- k eigenvectors and eigenvalues of the kernel matrix P
- ❑ Typical kernel functions:
 - ❑ (1) polynomial kernel: $\kappa(x_i, x_j) = (1 + x_i \cdot x_j)^p$
 - ❑ (2) radial basis function (RBF): $\kappa(x_i, x_j) = e^{-\frac{\|x_i - x_j\|^2}{2\sigma^2}}$
 - ❑ If we choose a linear kernel: $\kappa(x_i, x_j) = x_i \cdot x_j$, KPCA degenerates to the standard PCA
- ❑ Major formulas of Kernel PCA

Step 1: Proximity Construction

Step 2: Preserving Proximity

KPCA $P(i, j) = \kappa(x_i, x_j)$

$$\min \sum_{i,j=1}^n (P(i, j) - \hat{P}(i, j))^2 = \|P - \hat{P}\|_{fro}^2$$

Nonlinear Dimensionality Reduction (II): SNE

□ SNE (Stochastic neighborhood embedding)

- Construct a proximity matrix P using the formula: $P(i, j) = \frac{e^{-d_{ij}^2}}{\sum_{l=1, l \neq i}^n e^{-d_{il}^2}}$ where $d_{ij}^2 = \frac{\|x_i - x_j\|^2}{2\sigma^2}$
 - representing the probability that x_j is the neighbor of x_i
- Suppose we have learned the low-dimensional representations $\hat{x}_i$, we can compute another estimated proximity matrix in the similar way: $\hat{P}(i, j)$
- We want to make the estimated proximity matrix $\hat{P}$ to be as close as possible to P
- That is, we want to minimize the overall K-L divergence, that is,

$$\hat{x}_i = \arg \min_{\hat{x}_i, (i=1, \dots, n)} \sum_{i=1}^n D_{KL}(P_i || \hat{P}_i)$$

Step 1: Proximity Construction	Step 2: Preserving Proximity
--------------------------------	------------------------------

SNE	$P(i, j) = \frac{e^{-d_{ij}^2}}{\sum_{l=1, l \neq i}^n e^{-d_{il}^2}}$	$\min \sum_{i=1}^n KL(P_i \hat{P}_i)$
-----	--	--